

Metadata Privacy for FHE-Based Routing: A Formal Security Analysis

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<https://github.com/bon-cdp/f2chat>

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Abstract

We present a formal security analysis of metadata privacy in FHE-based messaging systems. While Fully Homomorphic Encryption protects message content, it does not inherently prevent traffic analysis attacks that can reveal communication patterns. We formalize the *unlinkability game* and propose an epoch-based identity rotation scheme with k -anonymity overlay. We prove that our scheme achieves “6-sigma” privacy ($\text{Adv} < 10^{-6}$) under standard cryptographic assumptions with concrete parameter choices.

1 Introduction

1.1 The Problem

The f2chat system uses Fully Homomorphic Encryption (FHE) to enable *blind routing*: the server routes messages homomorphically without ever decrypting them. Specifically, the server sees:

$$\{\text{Enc}(P_{\text{alice}}), \text{Enc}(P_{\text{bob}}), \text{Enc}(\text{message})\}$$

where P_u is user u ’s polynomial identity in the ring $\mathbb{Z}_p[x]/(x^n + 1)$.

Content Privacy: The FHE layer provides 128-bit security for content. Breaking $\text{Enc}(P)$ requires solving hard lattice problems (Ring-LWE).

Metadata Leakage: However, if Alice always uses the same polynomial identity P_{alice} , the server can observe:

- Same ciphertext structure appears at 9am, 12pm, 3pm daily
- Ciphertext B always receives from ciphertext A
- Frequency and timing patterns reveal the social graph

1.2 Our Contribution

We formalize the metadata privacy problem and propose a solution achieving:

1. **Unlinkability:** Different epochs use cryptographically independent identities
2. **k -Anonymity:** Within each epoch, users share anonymity sets
3. **6-Sigma Privacy:** Adversary advantage bounded by 10^{-6}

2 System Model

2.1 Polynomial Identities

Definition 2.1 (Polynomial Ring). *Let $p = 65537$ (prime) and $n = 64$ (degree). The polynomial ring is:*

$$R = \mathbb{Z}_p[x]/(x^n + 1)$$

All polynomial identities and messages are elements of R .

Definition 2.2 (Polynomial Identity Function). *A polynomial identity function is a mapping:*

$$P : \mathcal{U} \times \mathbb{Z}_{\geq 0} \rightarrow R$$

where $\mathcal{U}$ is the set of users and the second argument is the epoch number. We write $P_u(\varepsilon)$ for user u 's identity in epoch ε .

2.2 Epochs and Rotation

Definition 2.3 (Epoch Structure). *An epoch structure with duration Δ maps continuous time to discrete epochs:*

$$\varepsilon(t) = \lfloor t/\Delta \rfloor$$

The polynomial identity is constant within an epoch:

$$\varepsilon(t_1) = \varepsilon(t_2) \implies P_u(t_1) = P_u(t_2)$$

2.3 Server Observation Model

The server observes a sequence of tuples:

$$\mathcal{O} = \{(ct_{\text{sender},i}, ct_{\text{receiver},i}, ct_{\text{msg},i}, t_i)\}_{i=1}^q$$

where:

- $ct_{\text{sender},i} = \text{Enc}(P_{\text{sender}_i}(\varepsilon(t_i)), pk, r_i)$
- $ct_{\text{receiver},i} = \text{Enc}(P_{\text{receiver}_i}(\varepsilon(t_i)), pk, r'_i)$
- $ct_{\text{msg},i}$ is the encrypted message
- t_i is the timestamp
- r_i, r'_i are fresh randomness (BGV encryption is randomized)

2.4 Threat Model

Definition 2.4 (Honest-but-Curious Server). *The server:*

- *Observes all ciphertexts and timestamps*
- *Does not have access to any private keys*
- *Cannot decrypt any ciphertexts*
- *Cannot modify messages (integrity assumed)*
- *Attempts to link messages and infer the communication graph*

3 Security Definitions

3.1 The Unlinkability Game

Definition 3.1 (Unlinkability Game $\text{Unlink}_{\Pi,A}(\lambda)$). Let Π be an identity scheme and A a PPT adversary.

1. **Setup:** The challenger generates system parameters, including the PRF key.
2. **Observation Phase:** The adversary A observes q message tuples $\mathcal{O}$ as defined above, across multiple epochs.
3. **Challenge:** The challenger picks a random bit $b \in \{0, 1\}$:
 - If $b = 0$: Select two messages from the same user Alice at different epochs
 - If $b = 1$: Select two messages from different users Alice and Carol

The adversary sees only the ciphertexts and timestamps of these two messages.

4. **Guess:** The adversary outputs a guess $b' \in \{0, 1\}$.

5. **Output:** The game outputs 1 if $b' = b$.

The unlinkability advantage of A is:

$$\text{Adv}_{\Pi}^{\text{unlink}}(A) = \left| \Pr[\text{Unlink}_{\Pi,A}(\lambda) = 1] - \frac{1}{2} \right|$$

Definition 3.2 (Unlinkability Security). A scheme Π is (ϵ, q) -unlinkable if for all PPT adversaries A making at most q observations:

$$\text{Adv}_{\Pi}^{\text{unlink}}(A) \leq \epsilon$$

3.2 k -Overlap Property

Definition 3.3 (k -Overlap). Let $f : R \rightarrow \{1, \dots, K\}$ be a bucket function. An identity scheme has the k -overlap property if for all epochs ε and all users u :

$$|\{v \in \mathcal{U} : f(P_v(\varepsilon)) = f(P_u(\varepsilon))\}| \geq k$$

where $|\mathcal{U}|$ is the total number of active users.

Remark 3.4. The k -overlap property ensures that at any moment, a user is indistinguishable from at least $k - 1$ other users in the same “anonymity bucket.”

4 Proposed Scheme

4.1 Epoch-Based Deterministic Rotation

Definition 4.1 (PRF-Based Identity Derivation). Let $F : \{0, 1\}^\lambda \times \mathbb{Z}_{\geq 0} \rightarrow R$ be a secure PRF. For user u with seed s_u :

$$P_u(\varepsilon) = F(s_u, \varepsilon)$$

Key Insight: The seed s_u must be shared between Alice and her contacts. When Alice establishes contact with Bob, they exchange seeds (via QR code, secure channel, etc.):

- Alice learns s_{bob} and can compute $P_{\text{bob}}(\varepsilon)$ for any epoch
- Bob learns s_{alice} and can compute $P_{\text{alice}}(\varepsilon)$ for any epoch

Algorithm 1 Deterministic Identity Rotation

```
1: procedure GETIDENTITY( $s, t, \Delta$ )
2:    $\varepsilon \leftarrow \lfloor t/\Delta \rfloor$ 
3:   return  $F(s, \varepsilon)$  ▷ PRF evaluation
4: end procedure
```

4.2 k -Anonymity Overlay

To achieve the k -overlap property, we define a bucket function:

$$f(P) = H(P) \mod K$$

where $H : R \rightarrow \{0, 1\}^{256}$ is a hash function and $K = \lceil n/k \rceil$ for n users and target anonymity k .

Cover Traffic: To prevent frequency analysis, users generate dummy messages to their own mailbox at a constant rate, filling their bucket's expected capacity.

4.3 Seed Sharing Protocol

When Alice adds Bob as a contact:

1. Alice generates: $s_{\text{alice}} \xleftarrow{\$} \{0, 1\}^\lambda$
2. Exchange (via secure channel): Alice sends s_{alice} , Bob sends s_{bob}
3. Both store: ("Alice", s_{alice}), ("Bob", s_{bob})

Forward Secrecy Note: To limit damage from seed compromise, seeds can be ratcheted: $s^{(i+1)} = H(s^{(i)})$ with periodic re-keying.

5 Security Analysis

5.1 Main Theorem

Theorem 5.1 (Metadata Privacy). *Let Π be the epoch-based identity scheme with:*

- *PRF F with security parameter λ*
- *Epoch duration Δ*
- *k -overlap property with parameter k*
- *q total messages observed*

For any PPT adversary A :

$$\text{Adv}_{\Pi}^{\text{unlink}}(A) \leq \text{Adv}_F^{\text{prf}}(B) + \frac{q^2}{2^\lambda} + \frac{1}{k}$$

where B is a PRF adversary running in time $O(\text{time}(A))$.

Proof Sketch. We proceed via a sequence of games.

Game 0: The real unlinkability game.

Game 1: Replace $F(s_u, \cdot)$ with a truly random function $R_u(\cdot)$ for each user u . By the PRF security of F :

$$|\Pr[\text{Game 0} = 1] - \Pr[\text{Game 1} = 1]| \leq \text{Adv}_F^{\text{prf}}(B)$$

Game 2: In Game 1, identities in different epochs are truly random. The probability that two random polynomials collide is $1/|R| = 1/p^n$. By a union bound over q^2 pairs:

$$|\Pr[\text{Game 1} = 1] - \Pr[\text{Game 2} = 1]| \leq \frac{q^2}{p^n} \leq \frac{q^2}{2^\lambda}$$

Game 3: Given truly random, non-colliding identities, the adversary must distinguish same-user vs different-user based only on bucket membership. With k -overlap, the adversary's best strategy is guessing within the bucket:

$$\text{Adv}[\text{Game 3}] \leq \frac{1}{k}$$

Combining: $\text{Adv}_\Pi^{\text{unlink}}(A) \leq \text{Adv}_F^{\text{prf}}(B) + q^2/2^\lambda + 1/k$. □

5.2 Achieving 6-Sigma Privacy

Corollary 5.2 (Concrete Parameters for 6-Sigma). *For $\text{Adv} < 10^{-6}$ (6-sigma privacy) with:*

- $\lambda = 128$ (security parameter)
- $q = 10^6$ (messages observed)

We need:

1. **PRF term:** $\text{Adv}_F^{\text{prf}}(B) \approx 0$ for any reasonable adversary
2. **Collision term:** $q^2/2^\lambda = 10^{12}/2^{128} \approx 2.9 \times 10^{-27}$ (negligible)
3. **Anonymity term:** $1/k < 10^{-6}$ requires $k > 10^6$

Practical Relaxation: *If timing jitter provides $\sqrt{q}$ uncertainty:*

$$\text{Adv} \leq \frac{1}{k} \cdot \frac{1}{\sqrt{q}} = \frac{1}{k \cdot 10^3}$$

With $k = 1000$ and timing jitter over 1000 messages: $\text{Adv} \approx 10^{-6}$.

5.3 Attack Analysis

Lemma 5.3 (Attack 1: Ciphertext Linking). *BGV encryption is randomized: $\text{Enc}(P, pk, r_1) \neq \text{Enc}(P, pk, r_2)$ for $r_1 \neq r_2$. Thus ciphertext equality reveals nothing about plaintext equality.*

Lemma 5.4 (Attack 2: Timing Correlation). *With epoch duration Δ and message timing uniformly distributed within epochs, timing provides at most $\log_2(T/\Delta)$ bits of information, where T is the observation period. For $\Delta = 100\text{ms}$ and $T = 1$ year:*

$$\log_2(T/\Delta) \approx \log_2(3 \times 10^{11}) \approx 38 \text{ bits}$$

This is absorbed by the k -anonymity factor.

Lemma 5.5 (Attack 3: Frequency Analysis). *Cover traffic at rate ρ messages per epoch masks true communication patterns. With constant cover traffic, distinguishing real from cover requires breaking FHE.*

6 Practical Considerations

6.1 Epoch Duration Trade-offs

Epoch Δ	Advantage	Disadvantage
100ms	Minimal linkable window	High key derivation overhead
1 second	Good balance	Reasonable
1 minute	Low overhead	Larger linkable sets
1 hour	Very low overhead	More correlation possible

Recommendation: $\Delta = 1$ second provides a good balance.

6.2 Storage Requirements

For message retrieval across epochs, clients must store:

- Current seed: 128 bits per contact
- Epoch range for retrieval: 2 integers (16 bytes)
- No historical seeds needed if using PRF (can recompute)

Total: $O(c \cdot 32)$ bytes for c contacts.

6.3 Clock Synchronization

Clients must agree on epoch boundaries. With NTP synchronization to $\pm 50\text{ms}$ and $\Delta = 1\text{s}$, the probability of epoch disagreement is:

$$\Pr[\text{epoch mismatch}] = \frac{100\text{ms}}{1000\text{ms}} = 0.1$$

Mitigation: Accept messages from epochs $[\varepsilon - 1, \varepsilon + 1]$.

7 Conclusion

We have presented a formal security analysis of metadata privacy for FHE-based messaging. Our epoch-based identity rotation scheme with k -anonymity overlay achieves provable unlinkability with concrete parameters.

Key Results:

- Formal unlinkability game capturing metadata privacy
- Security reduction to PRF security
- 6-sigma privacy with $k = 10^6$ or $k = 1000$ with timing jitter
- Practical implementation in f2chat codebase

Future Work:

- Zero-knowledge proofs for routing validity
- Integration with Private Information Retrieval (PIR)
- Differential privacy for cover traffic optimization

A Implementation Roadmap

A.1 Phase 1: Extend PolynomialIdentity

```
// lib/crypto/polynomial_identity.h

class RotatingPolynomialIdentity {
public:
    // Create with PRF seed
    static absl::StatusOr<RotatingPolynomialIdentity>
        Create(const std::string& seed);

    // Get identity for specific epoch
    Polynomial GetIdentityForEpoch(int64_t epoch) const;

    // Current epoch from system time
    static int64_t CurrentEpoch(absl::Duration epoch_duration);

private:
    std::string seed_; // PRF seed (shared with contacts)

    // PRF: seed x epoch -> polynomial
    Polynomial PRF(int64_t epoch) const;
};
```

A.2 Phase 2: Anonymity Set Management

```
// lib/network/anonymity_set.h

class AnonymitySet {
public:
    // Compute bucket for identity
    static int64_t ComputeBucket(
        const Polynomial& identity, int64_t num_buckets);

    // Generate cover traffic
    static std::vector<EncryptedPolynomial> GenerateCoverTraffic(
        int64_t bucket,
        int count,
        const FHEContext& ctx);
};
```

A.3 Phase 3: Test Cases

```
// test/crypto/rotating_identity_test.cc

TEST(RotatingIdentityTest, DifferentEpochsUnlinkable) {
    auto id = RotatingPolynomialIdentity::Create(seed);
    auto p1 = id.GetIdentityForEpoch(0);
```

```

    auto p2 = id.GetIdentityForEpoch(1);

    // Statistical test: p1 and p2 should be independent
    EXPECT_NE(p1, p2);
    EXPECT_FALSE(AreCorrelated(p1, p2));
}

TEST(RotatingIdentityTest, SameEpochDeterministic) {
    auto id = RotatingPolynomialIdentity::Create(seed);
    auto p1 = id.GetIdentityForEpoch(42);
    auto p2 = id.GetIdentityForEpoch(42);

    EXPECT_EQ(p1, p2); // Same epoch, same identity
}

```